AWS EC2 TO RDS MYSQL

PHP Web Application with Classic Load Balancer, Route 53 and AWS Certificate Manager

FINAL PROJECT DOCUMENTATION
AWS Region: ap-south-1 (Mumbai)
Domain: vaibhavcloud.site
Prepared: 9 November 2026

1. Project Overview

This proof of concept demonstrates a PHP web application hosted on two Amazon EC2 instances and connected to Amazon RDS for MySQL. An internet-facing Classic Load Balancer distributes incoming requests across the registered EC2 instances. Amazon Route 53 provides DNS routing, and AWS Certificate Manager (ACM) supplies the public TLS certificate used by the load balancer.

Attribute	Project detail
Project name	AWS EC2 to RDS MySQL POC
Application	PHP form to insert and display user records
Compute	Two Amazon EC2 instances; Amazon Linux 2023; t3.micro
Web stack	Apache HTTP Server (httpd) and PHP
Load balancer	POC-CLB; internet-facing Classic Load Balancer
Database	Amazon RDS for MySQL; instance identifier database-1
DNS	Route 53 public hosted zone vaibhavcloud.site; www A/alias record
TLS	ACM-issued certificate for vaibhavcloud.site
Region	ap-south-1 (Mumbai)

2. Project Objectives

- Deploy a PHP web application on Amazon Linux 2023 EC2 instances.
- Distribute incoming application traffic through a Classic Load Balancer.
- Route the application domain to the load balancer using Route 53.
- Enable HTTPS using an ACM-issued certificate.
- Store and retrieve application records in Amazon RDS for MySQL.
- Validate EC2 health and load balancer target registration.

3. Architecture

The following is a conceptual reference diagram. The supplied AWS console evidence shows a Classic Load Balancer and both EC2 instances in ap-south-1a. The generated diagram labels an Application Load Balancer and depicts two Availability Zones; those details do not match the observed deployment. Treat Section 4 as the deployment source of truth and update the diagram before publishing it as an exact as-built diagram.

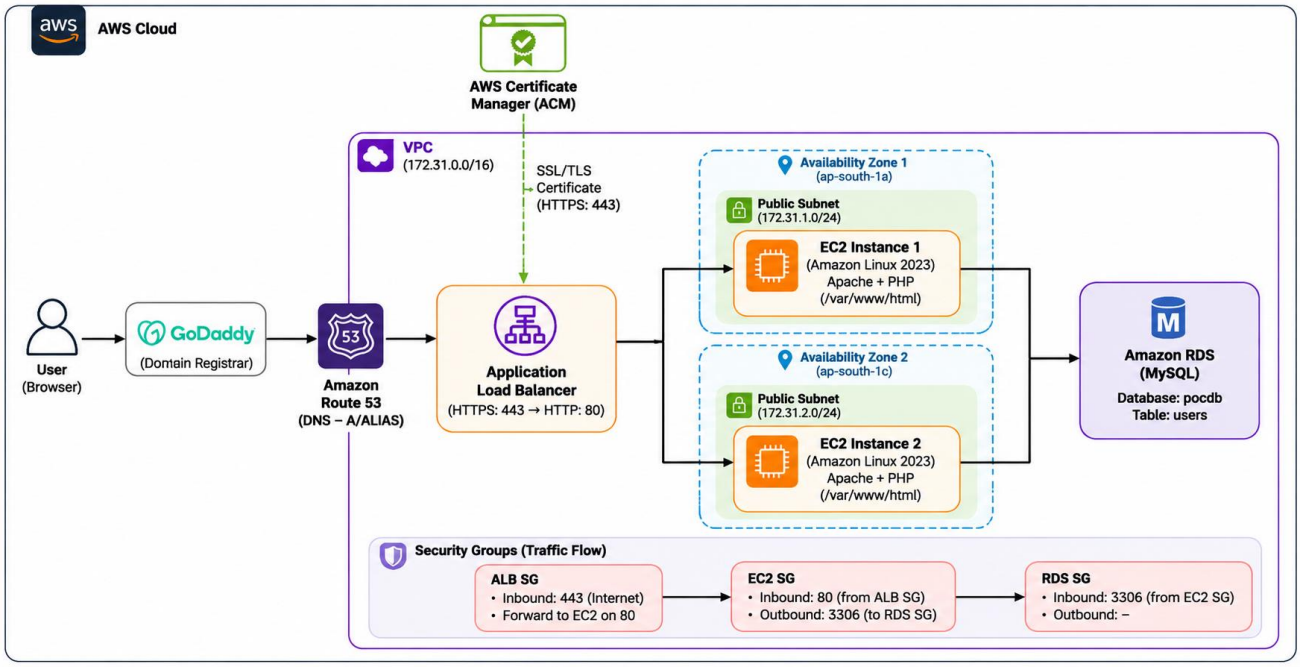


Figure 1. Conceptual architecture; verify load balancer type and AZ placement against the deployed environment.

4. Deployed AWS Resources

Resource	Observed configuration
VPC	Default VPC vpc-07810326fe51c5678; CIDR 172.31.0.0/16
Load balancer	POC-CLB; Classic; internet-facing
Load balancer DNS	POC-CLB-138794477.ap-south-1.elb.amazonaws.com
CLB subnets	Enabled subnets in ap-south-1a, ap-south-1b and ap-south-1c
EC2 instance 1	application server -1; i-06eecb766d2e92abc; t3.micro; ap-south-1a
EC2 instance 2	application server -2; i-0789c5bf1d82e6f13; t3.micro; ap-south-1a
Target health	2 of 2 instances In-service in supplied screenshot
RDS	database-1; MySQL Community 8.4.11; db.t4g.micro; ap-south-1c
RDS network	TCP 3306; Publicly accessible: No
RDS subnet group	default-vpc-07810326fe51c5678; three subnets shown
Route 53	Public zone vaibhavcloud.site; www A/alias to CLB
ACM	Amazon-issued certificate for vaibhavcloud.site; Issued; In use

5. Application Functionality

The PHP application provides Name, Email ID and Phone Number fields. On submission, it connects to MySQL, inserts a record into the users table and displays the latest records.

- Application database configuration: pocdb.
- Table: users.
- Columns: id, name, email_id, phone_no and created_at.
- The supplied PHP source creates the table if it does not exist.
- The application retrieves the latest 10 records, ordered by descending ID.

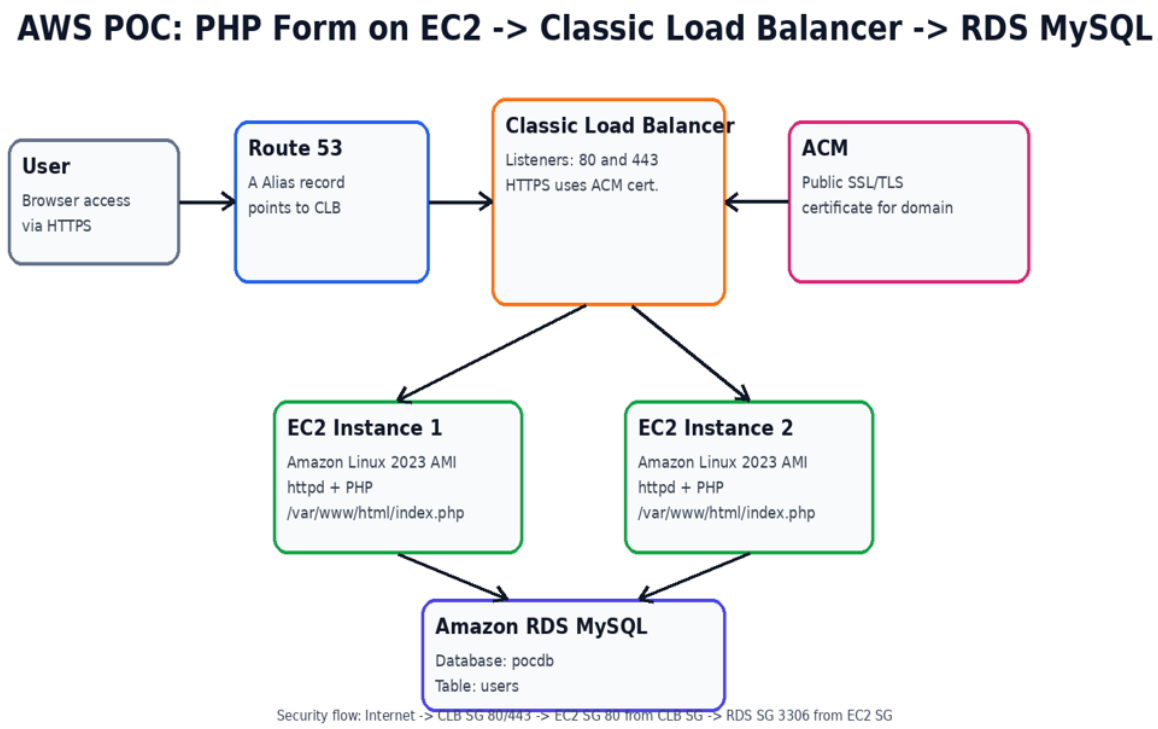


Figure 2. PHP form and records displayed from MySQL.

Configuration check: the RDS console screenshot displays the DB name as mysql, while the supplied PHP code uses pocdb. Confirm that pocdb exists on this RDS instance and that the deployed application uses the intended database.

6. Request and Data Flow

1. The user opens the configured domain in a browser using HTTPS.
2. Route 53 resolves the www record to the Classic Load Balancer alias target.
3. The load balancer accepts HTTPS using the ACM certificate.
4. The load balancer forwards application traffic to a registered EC2 instance over HTTP port 80.
5. Apache serves the PHP application from /var/www/html/index.php.
6. PHP connects to the private RDS endpoint over TCP port 3306.
7. MySQL stores submitted records in users and returns records for display.

7. Network and Security Design

Traffic path	Intended rule
Internet → CLB	Allow TCP 443 from intended public clients; allow TCP 80 only if redirect is configured.
CLB → EC2	EC2 security group allows TCP 80 from the load balancer security group.
EC2 → RDS	RDS security group allows TCP 3306 from the application EC2 security group.
Database exposure	Keep RDS private; screenshot shows Publicly accessible: No.
Administration	Restrict SSH TCP 22 to a trusted administrator IP or use Systems Manager.

Verification note: screenshots show default security groups attached to the load balancer targets and RDS. The exact rule lists were not provided. Verify the actual inbound and outbound rules in the EC2 and RDS security group pages before claiming the intended restrictions are implemented.

8. Deployment Summary

8.1 EC2 application servers

The servers use Amazon Linux 2023 with Apache and PHP. Typical package setup:

```
sudo dnf update -y

sudo dnf install -y httpd php php-mysqli mariadb105

sudo systemctl enable --now httpd

sudo systemctl status httpd
```

Deploy the same application code to /var/www/html/index.php on both EC2 instances. Configure database credentials securely outside the source repository.

8.2 RDS MySQL

The RDS instance database-1 runs MySQL Community 8.4.11 on db.t4g.micro in ap-south-1c. It is not publicly accessible and listens on port 3306.

8.3 Classic Load Balancer

POC-CLB is internet-facing. The supplied screenshot shows two registered application instances, both In-service. DNS name: POC-CLB-138794477.ap-south-1.elb.amazonaws.com.

8.4 Route 53 and ACM

The public hosted zone vaibhavcloud.site contains NS and SOA records and a www A/alias record pointing to the Classic Load Balancer. ACM shows an Amazon-issued certificate for vaibhavcloud.site with status Issued and In use. Confirm the certificate covers the exact hostname used by browsers.

9. Validation Evidence

Test	Observed result
EC2 health	Both instances Running; 3/3 status checks passed in supplied screenshot.
Load balancer targets	2 of 2 registered instances shown as In-service.
RDS availability	database-1 status shown as Available.
RDS exposure	Publicly accessible shown as No.
ACM	Certificate status Issued; In use shown as Yes.
Route 53	www A record shown as Alias to the CLB DNS name.
Application	Application screenshot displays three sample records (IDs 1–3).
Cross-AZ failover	Not evidenced; test and record before claiming high availability.

10. Security Improvements Before Publishing

- Rotate the database password previously placed directly in PHP source code or shared in chat; treat it as exposed.
- Use AWS Secrets Manager or Systems Manager Parameter Store; never commit secrets to GitHub.
- Create a least-privilege MySQL application user instead of using the RDS master account.
- Restrict EC2 HTTP ingress to the load balancer security group and RDS MySQL ingress to the EC2 security group.
- Use HTTPS and verify the certificate covers the exact application hostname.
- For new production deployments, prefer an Application Load Balancer; this project currently uses Classic Load Balancer.
- Place application instances in separate Availability Zones if cross-AZ resilience is a project goal, then test failover.
- Move schema creation into a controlled migration step instead of executing DDL on each web request.
- Use CSRF protection, input validation, prepared statements and output escaping in the PHP application.

11. Known Limitations and Accuracy Notes

- The deployed load balancer is Classic Load Balancer; the generated concept diagram labels it Application Load Balancer.
- Both EC2 instances are shown in ap-south-1a; the concept diagram depicts separate AZs.
- The supplied PHP code uses pocdb, while the RDS details screenshot displays mysql as the DB name. Verify the actual database.
- The exact security group rule lists were not supplied; confirm actual rules.
- The application screenshot shows records, but does not independently prove which EC2 instance served each request or prove load distribution.

12. Suggested GitHub Structure

```
AWS/
└── PROJECTS/
    └── EC2-RDS-MySQL-CLB-Route53-ACM/
        ├── README.md
        ├── .gitignore
        ├── application/
        │   └── index.php
        ├── architecture/
        │   └── aws-ec2-clb-rds-architecture.png
        ├── screenshots/
        │   ├── application-working.png
        │   ├── rds-configuration.png
        │   ├── rds-networking-configuration.png
        │   ├── acm-certificate-issued.png
        │   ├── route53-alias-record.png
        │   ├── clb-target-instances.png
        │   └── ec2-instances-running.png
        └── documentation/
            └── AWS_EC2_RDS_MySQL_POC_Final_Documentation.docx
```

Do not upload .pem files, passwords, environment files containing secrets, AWS access keys or private credentials. Use sanitized screenshots only.

13. Cleanup and Cost Management

When the POC is not in use, review and remove billable resources. A safe retirement process is to remove the Route 53 alias if retiring the application, delete the load balancer, stop or terminate EC2 instances, and take a final RDS snapshot before deleting the database if the data must be retained. Review hosted zones, ACM certificates, NAT Gateways, Elastic IP addresses and snapshots separately.

14. Conclusion

This POC demonstrates an AWS web application workflow using EC2, Apache, PHP, Classic Load Balancing, Route 53, ACM and Amazon RDS for MySQL. Supplied evidence confirms that both EC2 instances were healthy and registered with the load balancer, RDS was available and private, the ACM certificate was issued, and the PHP application displayed MySQL records. The limitations and verification notes identify the remaining checks needed to make the public portfolio documentation fully consistent with the deployed environment.

Appendix A. Final Publication Checklist

- ☐ Update architecture diagram to show Classic Load Balancer and actual AZ placement.
- ☐ Rotate exposed database password and configure credentials outside source code.
- ☐ Confirm pocdb exists and is the database used by the running application.
- ☐ Verify CLB listeners and certificate hostname coverage.
- ☐ Verify and capture CLB, EC2 and RDS security group rules.
- ☐ Test the application through the domain and record the results.
- ☐ Do not claim cross-AZ failover until configured and tested.